

Department of Computer Science

Project Name: AI Agent

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SUBJECT: Artificial Intelligence
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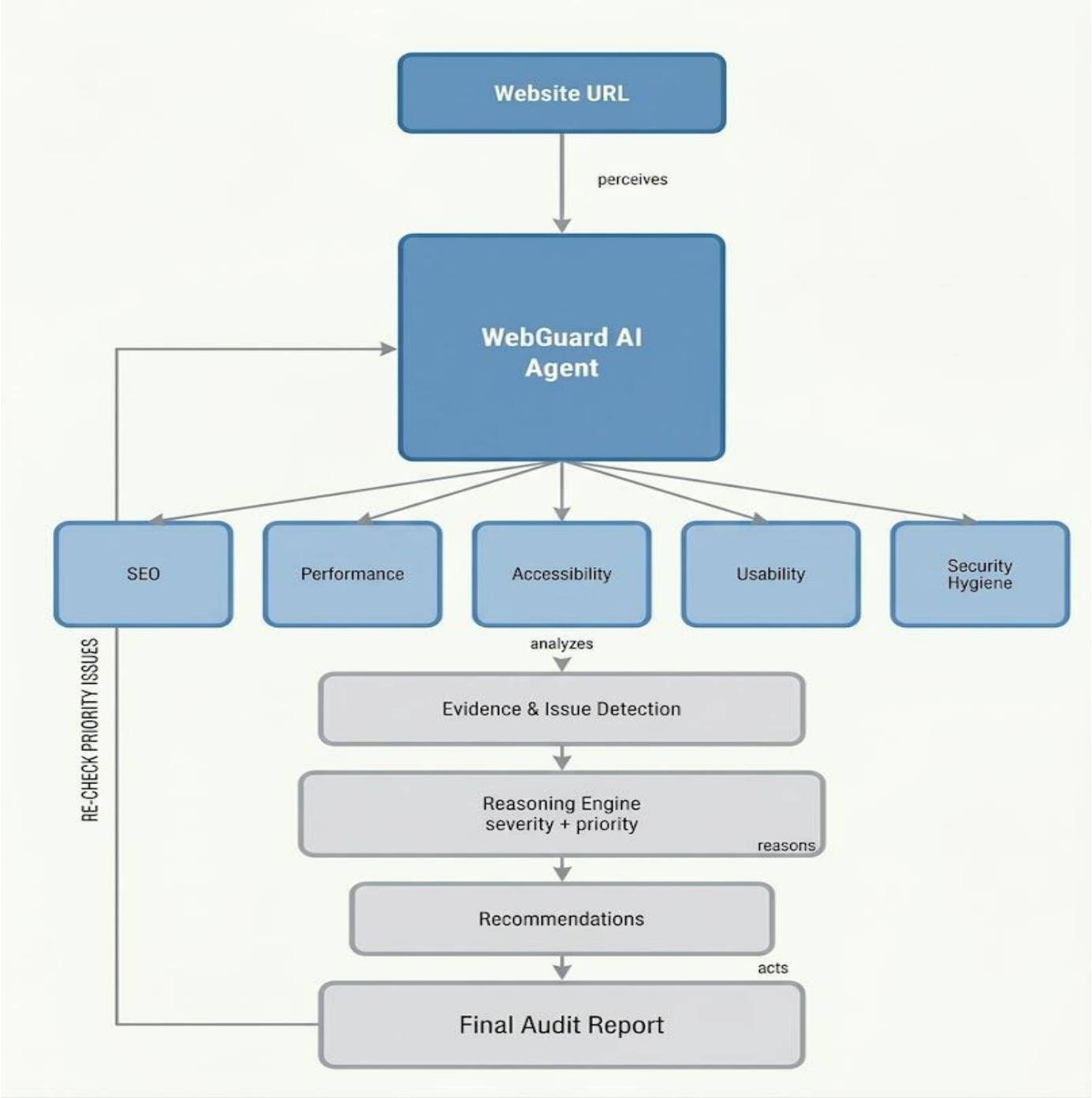
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WebGuard AI — *Autonomous Website Audit Agent*

PROBLEM FORMULATION:

Given a website URL, inspect the site and automatically identify SEO, accessibility, performance, usability, and basic security-hygiene issues, then return a prioritized report with recommendations.

GRAPHICAL REPRESENTATION



AGENT FORMULATION:

Environment: The target website — its pages, HTML, metadata, and linked resources — accessed over the public internet.

Sensors: URL, HTML structure, metadata, headings, images/alt text, links, load time, accessibility attributes, and HTTP responses.

Knowledge Base: SEO, accessibility, performance, and usability best practices, plus severity rules for ranking issues.

Algorithm: Collects data from the site, compares it against the knowledge base, flags deviations as issues, assigns severity, and ranks them by importance.

Actuator: Runs the relevant checks, classifies and prioritizes findings, generates recommendations, produces the report, and re-checks priority issues.

Example:

Issue	Evidence	Severity	Action
Missing page title	No <title> tag	High	Add a concise title tag.
Images without alt text	6 of 9 images missing alt	Medium	Add descriptive alt text.
Broken links	2 links return 404	High	Fix or remove broken links.
Slow page loading	Load time above benchmark	Medium	Compress images, enable caching.
Poor heading structure	h1 skips to h4	Low	Reorder headings sequentially.

ENVIRONMENT CLASSIFICATION:

Partially Observable: The agent can inspect the page's HTML and links, but not server-side details or login-only content.

Stochastic: The same site can return different results across audits due to network conditions or server load.

Sequential: Earlier findings affect later checks within the same audit — for example, an unreachable page limits what else can be checked.

Dynamic: A website's content and response time can change between or during audits, independent of the agent.

Discrete: Issues and severity levels are treated as fixed categories rather than continuous values.

Known: The rules used to judge a website (SEO, accessibility, performance guidelines) are defined in advance.

Single-agent: The agent audits a website on its own, without coordinating with other agents.

PEAS EVALUATION:

Performance Measure	Accuracy of detected issues, clarity of recommendations, and response time.
Environment	The target website — pages, HTML, metadata, links, and resources.
Actuator	Run checks, classify and prioritize issues, generate recommendations, produce report.
Sensor	URL, HTML structure, metadata, links, images/alt text, load time, HTTP responses.